

Unit 4

Elimination Pattern

- ***This pattern involves assessment of two areas:***

Urinary elimination and

Bowel elimination

- Application of nursing process

Subjective data on urinary elimination

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- Urinary habits (normal), color, amount.
- Changes in pattern (frequency, quantity)
- Problems such as pain, dribbling, hesitation, urgency, incontinence
- History of bladder surgery, catheterization
- Blood or pus in urine
- Effect of urinary problems on daily activities
- How the manages the problems of urinary elimination.

Objective data on urinary elimination

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- Inspection of urine
- Examination of lower abdomen (light palpation and percussion of bladder)

Subjective data on bowel elimination

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- Normal bowel movement pattern
- Recent changes in bowel movement pattern
- Color and consistency of stool
- Difficulty in maintaining bowel regularity (constipation, increased frequency)
- Actions taken to keep regularity (certain foods, enemas, laxatives, exercise)
- History of bowel surgery

Objective data on bowel elimination includes:

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- ☐ Inspection of feces
- ☐ Abdominal examination
- ☐ Rectal examination

Application of Nursing Process

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- ☐ Assessment
- ☐ Nursing Dx
- ☐ Planning
- ☐ Implementation
- ☐ Evaluation

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4.1

URINARY ELIMINATION

URINARY CATHETERIZATION

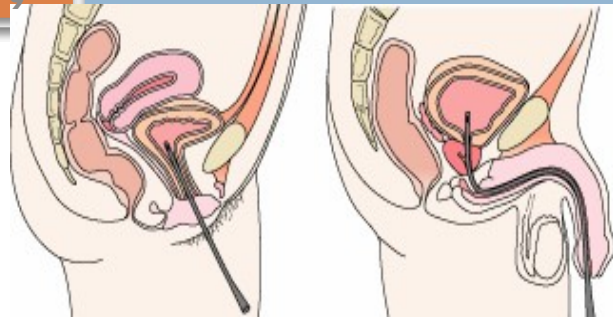
➤ **Categories of urethral catheters**

- ❑ Straight or Robinson catheter a single lumen tube i.e. commonly metallic types
- ❑ Retention or Foley catheter contains second lumen (two and three way catheter)

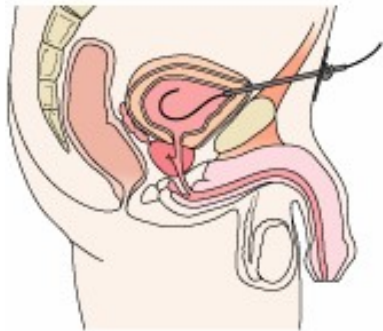
➤ **Types of Catheterization (routes)**

- ❑ Urinary catheterization
- ❑ Suprapubic catheterization

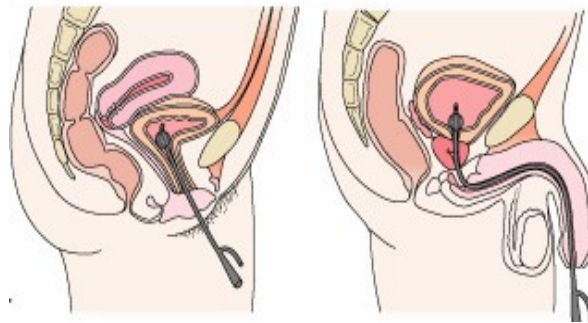
Catheters



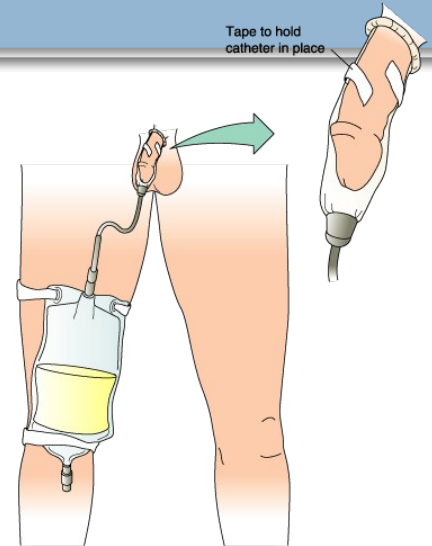
Straight



Suprapubic



Indwelling



Condom

- Selecting an appropriate catheter:
 - • May be made of
 - ⇐Plastic – for 1 week
 - ⇐Latex – 2-3 (rubber)
 - ⇐Silicon – for 2-3 month
 - ⇐Pelyvinylchloride (PVC) – 4-6
 - 1. Select the type of material in accordance with the estimated length of the catheterization period:

- 2. Determine appropriate catheter size
- - are determined by diameter of lumen
- - graded on French scale or number.
- • Catheter size depends on the size of the urethral canal
- ⇐# 8-10 Fr – children
- ⇐# 14-16 Fr – female adults, 12 Fr for young girls,
- ⇐# Average 18 Fr – adult male or (16-18 Fr)
- **NB. Fr= French Scale**

Cont'd...

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3. Determine appropriate catheter length by the clients gender

- For adult male – 40 cm catheter
- For adult females – 22 cm catheter

4. Select appropriate balloon size

- 5 ml – for adults
- 3 ml – for children but it is labeled on the catheters

☐ Catheterization Using a straight catheter

Purpose

- ☐ To relieve discomfort due to bladder distention
- ☐ To assess the residual urine
- ☐ To obtain a urine specimen
- ☐ To empty the bladder prior to surgery

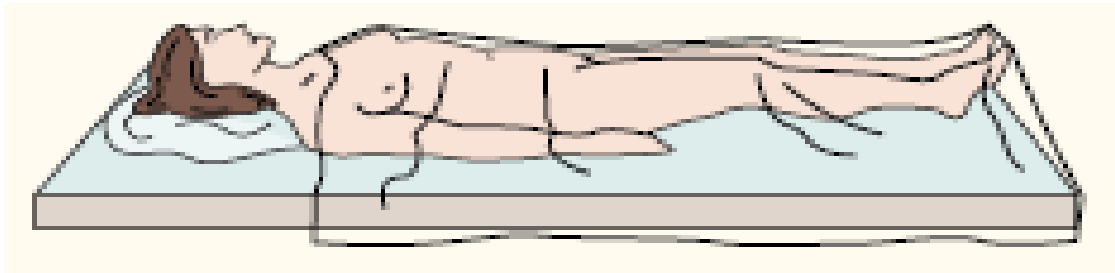
Position during catheterization procedure

- **Female:** dorsal recumbent (supine with knees flexed) or Sims position (side-lying with upper leg flexed at knee and hip)
- **Male:** supine position
 - Lubricate catheter 2.5 to 5 cm with K-Y jelly
 - Apply sterile drapes keep gloves sterile
 - Women: fenestrated over perineum
 - Men: over thighs and fenestrated over penis

Supine position

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- Supine position is putting patient in back lying often with a small pillow to support the head and shoulder



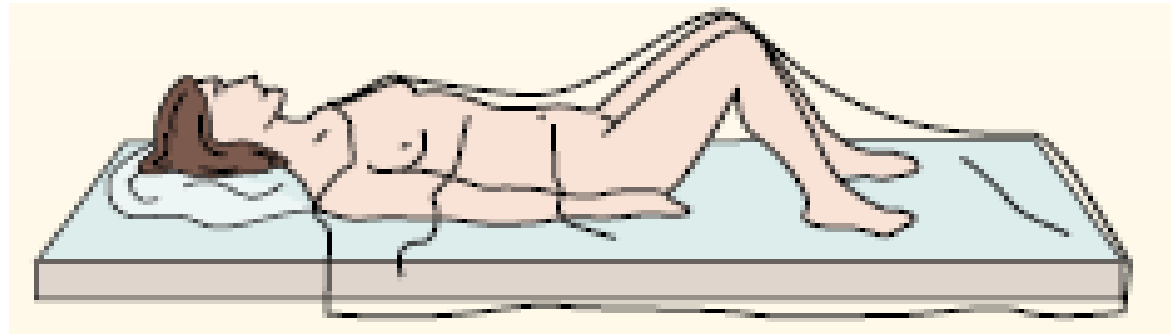
Purpose

- Promote comfort
- Male catheterization
- To help healing after certain abdominal operations

Dorsal recumbent

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- Is putting patient in back lying position with knees are flexed and the soles of the feet flat on the bed



Purpose

- Promote comfort
- For visualize the perineum
- To insert urinary catheter
- To relief pressure from ileum, knee and ankle

Cont'd....

- Place sterile tray and contents between legs
- Cleanse meatus:

Women: with non dominant hand, expose meatus, maintain Position of hand, cleanse with forceps, wipe from front to back, new cotton ball each swipe, far labial fold, near, and directly over meatus

Men: retract foreskin, hold penis below glans, maintain position of hand, with forceps clean in a circular motion from meatus down to base of glans, repeat three more times

Cont'd....

Hold end of catheter loosely coiled in dominant hand, place end of catheter in tray

Insert catheter:

Women: ask client to bear down as if to void, insert 5 to 7.5 cm or until urine flows, then advance another 2.5 to 5 cm

Men: hold penis perpendicular, ask client to bear down, insert 17 to 22.5 cm or until urine flows, then advance 2.5 to 5cm to insure in the bladder

Cont'd....

- Collect specimen if indicated
- Inflate balloon with amount indicated
- If client complains of pain, aspirate solution and advance catheter further and inflate
- Gently pull to feel resistance
- Attach catheter to collection bag and attach to bed frame below bladder
- Allow bladder to empty, 500 to 1 000 ml may void at a time
- Anchor catheter (thigh if appropriate and coil tubing on bed and attach to mattress)

Evaluate

- Palpate bladder
- Assess comfort
- Characteristics and amount of urine
- N.B. If you use three way catheter only for voiding, other than bladder irrigation, close one tip by needle cap

Document

- Report and record type and size of catheter
- Amount of fluid used to inflate balloon
- Characteristics of urine, amount, reason for catheter, specimens, client's response

Cont'd....

Urine color

- Normal urine color: golden yellow
- Reddish discoloration: shows RBCs or degradation products of RBC are present in the urine.
- Tea colored urine: shows presence of urobilinogen in the urine.
- Watery urine: shows excessive diuresis or hydration.
- Whisky colored: shows concentrated urine with dehydration.

Lower UTI symptoms

- **Frequency:** urinating again in less than 2 hrs after finishing voiding.
- **Urgency:** difficulty to postpone voiding.
- **Hesitancy:** difficulty to start urination; may be associated with straining.
- Poor (weak) stream of urine

- Intermittency: stopping and starting urination in one episode of voiding.
- Sense of incomplete emptying after finishing
- Dysuria: pain before, during or finishing micturation.
- Terminal dribbling: difficulty in completing micturation in a 'clean stop' fashion.

Cont'd....

Control of voiding

- For voluntary control of micturation, the body needs:
 - **CNS:** for the psychic stimulation, onset of stream and modulation of autonomic signals.
 - **ANS:** for detrusor activity modulation
 - Detrusor mechanism
 - Sphincter mechanism

Incontinence

- Involuntary leakage of urine through the urethral meatus.
- Can be:
 - Stress incontinence
 - Urge incontinence
 - Overflow incontinence
 - Neurogenic incontinence

Urinary catheterization

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- Involves inserting a small tube/ catheter through the urethra in to the bladder to allow urine to drain
- **Catheters:** are tubes commonly made of rubber or plastics, although certain types are made of woven silk or metal.

Purpose

- To empty the bladder in case of retention of urine, occurring as a post operative complication
- In case of retention due to injury or tumor of the spinal cord
- In case of obstruction due to the blockage of the urethra causing stricture
- To obtain sterile specimen of urine

Cont'd...

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- To ensure that the bladder is empty before an abdominal or pelvic operation or Paracentesis
- To keep incontinent patient dry
- To avoid contamination after operation of vagina or perineum
- To empty bladder irrigation or instillation of the bladder
- To determine if residual urine is present in the bladder
- For an accurate measurement of urinary out put
- To facilitate healing of urethra

Bladder Irrigation

- It is the washing out of the bladder to clear the catheter and/or the bladder

➤ Purpose

- To clean the bladder before operation depending on the surgeons order
- To arrest bleeding from the bladder
- To clean the catheter from mucous or blood clots
- To clean bladder form pus

Suprapubic catheter

- Suprapubic catheter is inserted through the abdominal wall above the symphysis pubis into the urinary bladder.
- **Purpose**
 - To prevent bladder infection
 - To keep skin integrity

Care of clients with Suprapubic catheter include

- Regular assessment of the client's urine, fluid drainage system.
- Skin care around the insertion site involves sterile technique.
- Periodic clamping of the catheter preparatory to removing it and measurement of residual urine.

- Leaving the catheter open to drainage for 48 to 72 hours then clamping the catheter for 3 to 4 hour periods during the day the client can void satisfactory amounts.
- Dressing should be changed whenever they are soiled
- A small amount of iodine is used.

BOWEL ELIMINATION

Gastric aspiration

- Aspiration is the withdrawal of fluid or gas from gastric cavity by sectioning.
- Gastric content analysis is examination of the contents of the stomach, primarily to determine the quantity of acid present and incidentally to ascertain the presence of blood, bile, bacteria, and abnormal cells.

Purpose

- ❑ To relieve stomach or intestinal distension following abdominal surgery.
- ❑ In case of gastrointestinal obstruction to remove the stomach content,
- ❑ To keep the stomach empty before an emergency abdominal operation is done.
- ❑ To aspirate the stomach contents for diagnostic purpose ,like detect acid-fast bacillus in a client with undiagnosed tuberculosis, total absence of hydrochloric acid is diagnostic of pernicious anaemia

Gastric aspiration.....

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Two types of gastric aspiration

- ***Intermittent method:*** -
 - Suction is done as condition required & as ordered.
- ***Continuous method:*** -
 - Attached to the drainage bag.

Two ways of suctioning:

- **A Simple suction**
 - By use of syringe.
- **B An electrical suction machine**
 - By use of an electrical suction device

Gastric aspiration.....

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- Clean large tray
- NGT
- Aspiration syringe
- Suction machine
- Gallipot with lubricant (liquid paraffin or vaselline) to lubricate the nostril.
- Saline solution or plain water in gallipot incase the stomach content is too thick to cone out through the syringe.
- Rubber mackintosh & towel to prevent pt's chest.
- Receiver for soiled swabs.
- Sodium bicarbonate or saline solution to clean the nostril.
- Water in kidney dish to test right position of the tube.
- Test tube & laboratory formats.

Gastric aspiration.....

□ Procedure

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- Explain the procedure to the client.
- Wash your hand, Assemble all the necessary material.
- Prop up an upright position with back rest or pillow.
- Cleanse & lubricate the nostrils.
- Lubricate the NGT with water.
- Insert the tube according to the procedure.
- Instruct the pt to open his mouth & make sure that the tube is in the stomach.
- Inject about 15 to 20cc of saline or water in to the stomach.
- With draw the fluid with syringe or suction & collect specimen if needed.
- If NGT is to be left in site then spigot or clamp is used to close end but if it is for one aspiration, remove immediately & very gently.

N.B: Prevent mouth & nose from dryness, always measure the amount, color, content, & smell for the with drawal solution.

- ♦ Report any change to the pt condition regarding: *PR, Temp., BP and fluid out put*

Gastric Lavage

- Gastric lavage is the introduction of solution into the stomach and removing gastric contents through NGT for washing out the stomach.
- **Purpose**
 - ❑ To remove inserted poison, other than corrosive substances like ammonia and mineral substances.
 - ❑ To introduce ice water or normal saline solution in tackling bleeding.
 - ❑ To cleanse the stomach before operation.
 - ❑ For diagnostic purposes.
 - ❑ To relief congestion, nausea and vomiting .

Indication

- Pyloric stenosis Or GOO
- Poisoning
- Preoperative care
- UGI bleeding
- Aspiration of gastric content for diagnosis
- To provide radiologic contrast to GI

❖ As a Treatment (Rx) Purpose

- Gastric decompression
- Feeding (enteral nutrition)
- Administration of medications
- Fluid and electrolyte therapy
- Bowel irrigation (stomach lavage)

□ Position

- Left lateral for unconscious pt
- Semi fowler's for conscious pt

➤ **Size of the NG - tube $16+(age/2)$ in French**

Gastric lavage.....

41 **Contraindications**

- ☐ After acid or alkaline ingestion
- ☐ “ Corrosive agents
- ☐ “ Ingestion of hydrocarbon (kerosene, gasoline)

Equipment

- ☐ NGT (nasogastric tube)
- ☐ Large irrigating syringe
- ☐ Plastic fannel
- ☐ Lavage solution
- ☐ Pail, News paper

Procedure

- Measure from the tip of the nose to ear lobe and distend to xiphoid process then mark the tube with plaster
- Lubricate the tube with cold water
- During gastric lavage the patient is positioned on his left side to allow pooling of the gastric content and decrease the passage of fluid into the duodenum during lavage
- Pass the NGT with advance downward force and give a moment rest when it reaches to pharynx
- As aspirate the stomach contents with syringe {funnel} attached to the tube before instilling lavage solution and save specimen.

Cont'd...

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- Attach the large syringe (funnel) to the end the tube an inject lavage solution in to NGT
- Elevate the fannel above the patients head and pour approximate 500 ml (total 2 liters)
- Repeat the procedure until the return is clear
- In still charcoal mixed with normal saline to remain in stomach (this reduces absorption)
- Pinch the tube during removal to prevent aspiration
- Give mouth care

ENEMA

- Is an injection of a liquid in to the rectum, to be returned or retained.
- The term enema is used to refer to the process of instilling fluid through the anal sphincter into the rectum and lower intestine for a therapeutic purpose

➤ **Purpose:**

- ❑ To cleanse the lower bowel,
- ❑ To assist in the evacuation of stool or flatus
- ❑ To instill medication

❖ **Cleansing enema/evacuating enema/**

- is n enema is the cleansing of a portion of the bowel by insertion of fluid rectally

Purpose

- To relieve gas, constipation or fecal impaction
- To cleanse the bowel prior to surgery, childbirth, or diagnostic examination.
- To evacuate the bowel in patients with neurologic dysfunction.
- Evacuates feces in clients with hemiplegia, quadriplegia or paraplegia
- Delivers medication

- **Types of liquid used for cleansing enema**
- **Tape water** = 500 to 1000 cc
- **Soap solution** = 500 to 1000 cc
- **Normal saline** - made by mixing one teaspoon of salt in a liter of water usually contains 1000cc of normal saline.
- **Epsom salt** 15 gm - 120 gm in 1000 cc of water

Cont'd...

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- No need to use too much soap - this may produce sever irritation of the membrane of the colon.
- Tap water must be administered consciously for infants or adults who have altered kidney or cardiac reserve this is to avoid water intoxication.
- Fill water container with 750 to 1 000cc of Luke warm solution

➤ **Contraindications**

- ☐ Rectal surgery
- ☐ Rectal/anal cancer
- ☐ Rectal infection
- ☐ For a patient with appendicitis
- ☐ Rectal/anal fissure

RETENTION ENEMA

- it is the injection of a liquid in to the rectum, to be retained in the rectum for some period of time (1 to 3hrs)

➤ **Purpose:**

- To supply the body with fluid
- To give medication
- To soften impacted fecal matter

RECTAL WASHOUT

- Is the injection of a liquid in to the rectum to be wash out the rectum and colon
- The tube is inserted into the rectum 7 to 10 cm

➤ Purpose

- To prepare the patient for x-ray examination and sigmoidoscopy
- To prepare the patient for rectum and colon operation

➤ **Precaution**

- ❑ The rectal wash out should not exceed for more than 2 hours
- ❑ The rectal wash out should be finished one hour before examination (e.g. x-ray, sigmoidoscopy) this is to give time for the large intestine to absorb the rest of the fluid
- ❑ Give cleansing enema half hour before the rectal wash out.

Inserting a rectal tube

- The insertion of a rectal tube is done to manage flatulence (gas) following abdominal surgery and/or reduce abdominal distention due to flatulence.
- Insert the tube 10 to 15 cm into the rectum

➤ Purpose

- It can be used to alleviate abdominal distention.
- It is used to control diarrhea that cannot be controlled with medical management and/or the use of rectal pouches, pads, or diapers due to extensive skin breakdown

Colostomy

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- Is a surgical opening by which an artificial opening is made in to the colon.

Purpose-

- To allow drainage of bowel content (as one type of fecal diversion) when there is an obstruction of the large bowel or a known lesion, such as cancer, that will cause an obstruction.
- To divert the fecal stream from the affected area to allow healing of the affected area of the bowel distal to it.
- It is also done as a palliative measure as a treatment of an obstruction caused by an inoperable growth of the colon, or if the rectum must be removed to treat cancer.

Colostomy...

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- To provide a permanent means of bowel evacuation
- Colostomy can be temporary or permanent
- A temporary colostomy allows the bowel to rest and later may be reanastomosed
- This procedure can also be used to treat inoperable bowel cancer with the ostoma placed proximal to cancer
- It is most commonly placed at the mid point of colon or transverse colon but permanent colostomy is placed at the sigmoid colon.

Colostomy Care

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- Postoperative nursing care should focus on assessing the stoma, protecting the skin around the stoma, selecting appropriate pouch and assisting the pt to adapt to the changed body psychologically.
- Normally the color of the stoma should be pink
- A dusky blue color indicates ischemia, and brown black color of stoma indicates necrosis
- So the nurse should assess the stoma color every 8hrs.
- There is mild to moderate swelling of the stoma the first 2 – 3 hrs after surgery.

Colostomy care...

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Generally:

- The skin around the stoma should be washed with mild soap rinsed with warm water, dried thoroughly, and then barrier should be applied to protect the skin.
- The volume, color and consistency of the drainage are recorded every time the pouch is changed.
- A pouch should never be placed on an irritated skin without a protective placed in between the pouch and the pts skin.

Colostomy irrigation

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Purpose

- To encourage bowel motion in a recently established colostomy and to ensure that the opening is patent.
- To relieve constipation in pts who has difficulty in managing their colostomy.
- To teach the how to establish regularity of evacuation through the colostomy.
- To reduce distension before closure of colostomy.

Colostomy irrigation....

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Equipment

- Irrigation can
- Tubing clip
- Rectal catheter, two large receivers
- About 500ml solution warmed at a 40 degree centigrade
- Colostomy bag, bed protective equipment
- IV stands, screen.

Procedure

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- ☐ Explain the procedure.
- ☐ Assemble all the necessary equipment and bring to bedside.
- ☐ Screen the pt and remove the dressing.
- ☐ Position the pt in sitting position and expose the abdomen.
- ☐ Place the mackintosh and draw sheet.
- ☐ Fill the irrigation can with the solution and elevate the can about 50cm above, remove air and clump it.
- ☐ Place the bucket on the floor beside the bed.
- ☐ Place large receiver below the colostomy.
- ☐ Connect the catheter with the tube to the irrigating can and lubricate it.

Colostomy care....

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Insert the catheter 6 - 8cm in to the colon.

- Allow the solution to run very slowly and remove the catheter for the return flow.
- Repeat the procedure until the colon is clearer.
- Give the pt clean abdominal pad to hold over colostomy during expulsion of gas.
- With draw the tube and leave the large tube as long as necessary.
- While the solution is being expelled remove the equipment.
- Return to the pt at intervals to assist if needed.
- Clean the area with soap and water (depending on the condition and duration of the colostomy), apply Vaseline gauze to prevent irritation, and apply complete dressing.
- .

Colostomy care...

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Provide good ventilation after the procedure.

- Record the procedure.
- **N.B:** Give the irrigation soon after the breakfast in order to establish regularity of evacuation through the colostomy.
 - ▣ If the area around the colostomy is excessively irritated apply ointments.
 - ▣ If the colostomy is permanent, the pt should be thought as early as possible.
 - ▣ Colostomy irrigation can also be made in the toilet allowing the return drainage to flow in to the toilet having the pt sit on the chair or on the toilet if it is sitting type



10Q FOR U'r
ATTENTION!!

